Unit 13, Lesson 12: Equivalent Expressions Using Properties of Operations

Benchmark

MA.7.AR.1.2 Determine whether two linear expressions are equivalent.

Learning Targets

- ☐ I can use properties of operations to write equivalent forms of an algebraic expression.
- ☐ I can determine whether two linear expressions are equivalent.

13.12.1 Warm-Up: Fact or Fiction?

1. Determine whether each equation below is fact (true) or fiction (false).

Equation	Fact or Fiction
$(30 + 8) \cdot 9 = 270 + 72$	○ Fact ○ Fiction
$4 + 5(9 - 15) = 9(9 - 15)$	○ Fact ○ Fiction
$2 \cdot (7 + 8) = (2 \cdot 7) + 8$	○ Fact ○ Fiction
$(5 + 4) - 10 = 5 + (4 - 10)$	○ Fact ○ Fiction

13.12.2 Refresher: Rewriting Expressions Using Properties of Operations

1. Fill in the missing blanks for each property.

   a. Using the ***commutative property of addition***, $3 + 12$ can be rewritten as ________________. Both expressions have a sum of ______________.

   b. Using the ***commutative property of addition***, $4x + 5$ can be rewritten as ______________. Both expressions will have the same sum regardless of the value of x.

   c. Using the ***commutative property of multiplication***, $(4)(-2)$ can be rewritten as _____________. Both expressions have a product of ____________.

   d. Using the ***commutative property of multiplication***, $(-3y)(0.5)$ can be rewritten as _______________. Both expressions will have the same product regardless of the value of y.

   e. Using the ***associative property of addition***, $-5 + (6 + 1)$ can be rewritten as _____________. Both sums are equal to ______________.

   f. Using the ***associative property of addition***, $9x + (-2x + 5)$ can be rewritten as _____________. Both expressions will have the same sum regardless of the value of x.

   g. Using the ***associative property of multiplication***, $2 \cdot (7 \cdot 3)$ can be rewritten as _____________. Both products are equal to _____________.

   h. Using the ***associative property of multiplication***, $(-6)(4y)$ can be rewritten as _____________. Both expressions will have the same product regardless of the value of y.

   i. Using the ***distributive property of multiplication over addition***, $5(6 + 2)$ can be rewritten as _____________. Both expressions have a value of ______________.

   j. Using the ***distributive property of multiplication over addition***, $7(3y + 8)$ can be rewritten as _____________. Both expressions will have the same product regardless of the value of y.

2. Discuss with your partner why the associative property and the commutative property do not hold true for subtraction and division. Give an example of each to show mathematically that these properties do not apply to subtraction and division.

3. Jay gave the example $6 + {}^{-}11$ is equal to ${}^{-}11 + 6$ to show that subtraction is commutative. Explain to Jay why his example does not show that subtraction is commutative.

13.12.3 Guided Instruction: Equivalent Expressions Using Properties of Operations

Algebraic expressions are considered equivalent if when evaluating the expression by substituting values in for variables, the resulting values are the same.

Algebraic expressions are also considered equivalent when one expression is rewritten using one or more of the properties of operations.

1. Complete each statement.

   a. The
      - ○ associative
      - ○ commutative
      - ○ distributive

      property can be used to state that $5(x+2)-3x$ is equivalent to $5x+10-3x$.

   b. The
      - ○ associative
      - ○ commutative
      - ○ distributive

      property can be used to state that $\frac{2}{3}x + \left(\frac{3}{8} + \frac{5}{6}x\right)$ is equivalent to $\frac{2}{3}x + \left(\frac{5}{6}x + \frac{3}{8}\right)$.

2. Which properties can be used to show that $2(x-4)+5(3+6x)$ is equivalent to $2x+30x-8+15$?

13.12.4 Guided Practice: Rewriting Expressions Using Properties of Operations

1. Write an equivalent expression using a property of operations. State the property used.

Expression	Equivalent Expression	Property Used
$-6(2-3x)+1-\frac{x}{2}$		
$12x+10+4x-17$		
$5x+(3x+9)+2$		

2. Several expressions are given below. Determine which expressions are equivalent using properties of operations.

$35x+(7x+56)$	$5(8x+7)$	$35x+56$
$(8x+7)\cdot 5$	$7(5x+8)$	$(35x+7x)+56$

Complete the table by writing the expressions that are equivalent based on the property used.

Equivalent Expressions		Property Used
		Associative Property
		Commutative Property
		Distributive Property

13.12.5 Your Turn!

Write an equivalent expression using the associative, commutative, or distributive property. State the property used.

1. $10x+\frac{2}{5}(-30+15x)$

2. $(18x+15)+4x-17$

3. $\left(-\frac{1}{5}x+6\right)\cdot 5$

4. $\frac{5}{4}n + \frac{7}{3} - \frac{11}{2}n - 12\frac{1}{3}$

5. $-3\left(\frac{1}{2}x + 4\right) - \frac{3}{5}(-1 + 10x)$

13.12.6 Wrap-Up: Ticket Out the Door

Use properties of operations to determine if the expressions are equivalent.

1. $\frac{4}{5}(x - 1) - \frac{1}{8}(2x + 1)$ and $\frac{11}{20}x - \frac{37}{40}$

2. $1\frac{3}{10}x + \frac{1}{5}x - \frac{1}{10} - x - \frac{1}{10}$ and $1\frac{1}{10} - \frac{4}{5}x$